

Experiment – 1.1

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Subject Name: Technical Training-II

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Section/Group: 25MCA-2A

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Aim

To design a responsive and interactive web interface for an Online Grocery Delivery System using HTML, CSS, Bootstrap, and JavaScript.

Procedure

Step 1: Requirement Analysis

- Identify user requirements.
- Decide the modules such as Home, Categories, Offers, Cart, Product Details, and Contact.

Step 2: Design

- Design the homepage layout.
- Create ER Diagram.
- Create DFD (Level 0 and Level 1).
- Design responsive UI using Bootstrap.

Step 3: Development

- Create HTML pages.
- Apply CSS for styling.
- Use Bootstrap components.
- Implement JavaScript for search, filtering, and shopping cart.

Step 4: Testing

- Check navigation.
- Test filters.
- Test Add to Cart.
- Verify responsive design on mobile, tablet, and desktop.

Step 5: Deployment

- Store project files.
- Open `index.html` in a browser or deploy on GitHub Pages/Netlify.

ER and DFD Diagram

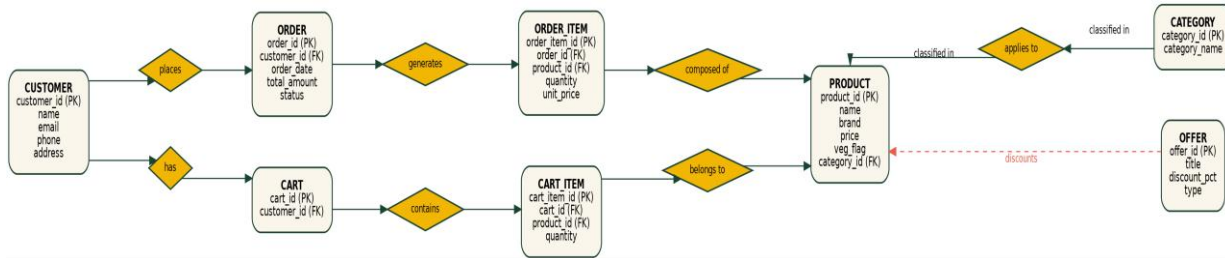


Figure 1 — ER diagram for the Online Grocery Delivery System

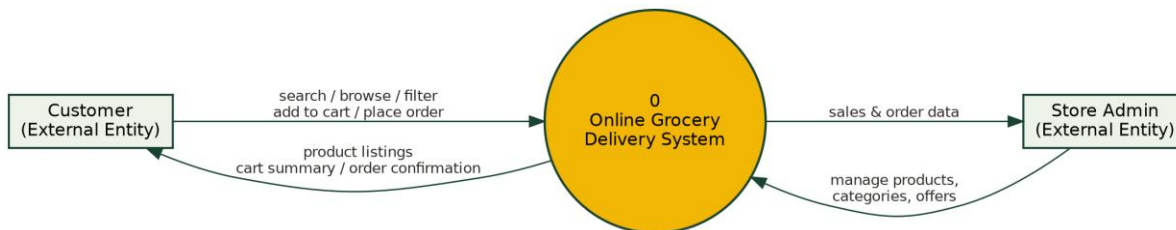


Figure 2 — DFD Level 0 (Context Diagram)

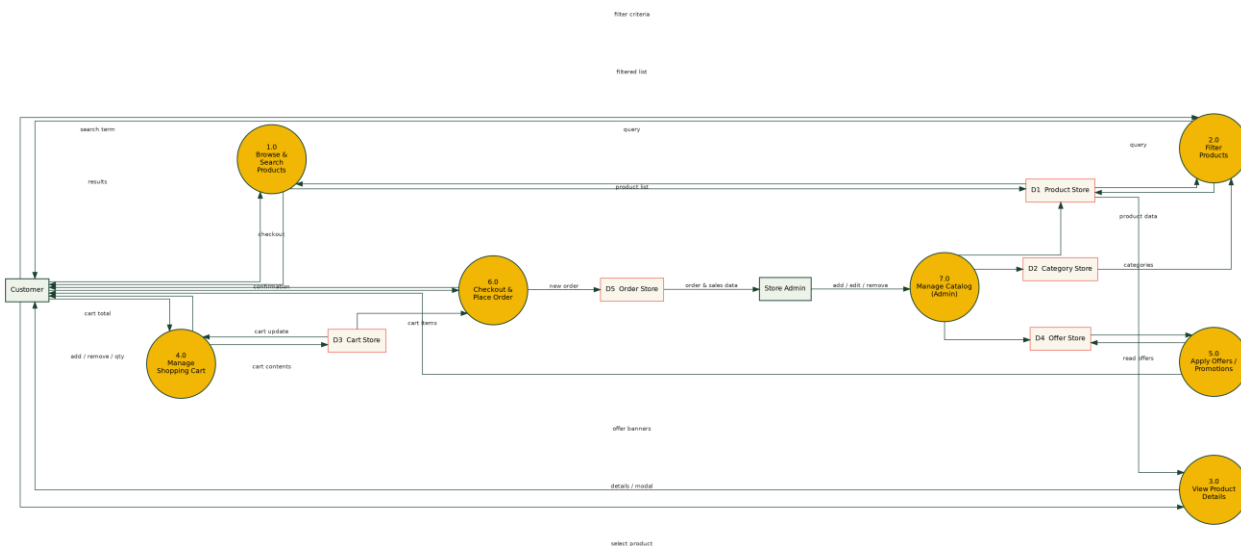


Figure 3 — DFD Level 1

Code

Navigation Bar Module

```
nav class="navbar navbar-expand-lg navbar-harvestly sticky-top">
  <div class="container">
    <a class="navbar-brand brand-mark" href="#home">harvestly<span class="dot">.</span></a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#mainNav"
    style="border-color: rgba(251,246,236,.4)">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="mainNav">
```

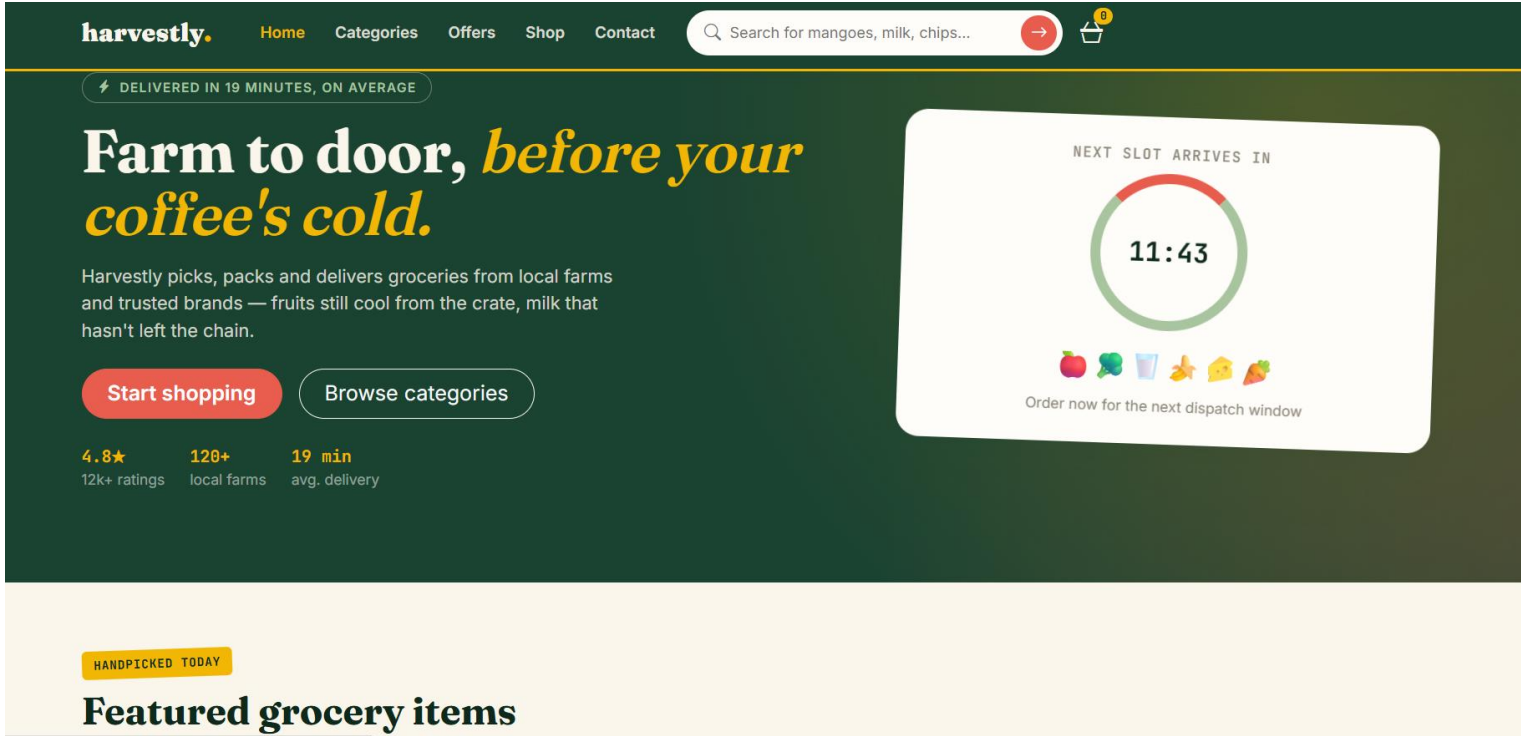
```
<ul class="navbar-nav mx-lg-3">
  <li class="nav-item"><a class="nav-link active" href="#home">Home</a></li>
  <li class="nav-item"><a class="nav-link" href="#categories">Categories</a></li>
  <li class="nav-item"><a class="nav-link" href="#offers">Offers</a></li>
  <li class="nav-item"><a class="nav-link" href="#shop">Shop</a></li>
  <li class="nav-item"><a class="nav-link" href="#contact">Contact</a></li>
</ul>
<form class="receipt-search flex-grow-1 my-3 my-lg-0 me-lg-3" style="max-width:360px" onsubmit="return
runSearch(event)">
  <i class="bi bi-search text-secondary"></i>
  <input id="searchInput" type="search" placeholder="Search for mangoes, milk, chips..." aria-label="Search
groceries">
  <button type="submit" aria-label="Search"><i class="bi bi-arrow-right"></i></button>
</form>
<div class="d-flex align-items-center gap-3">
  <a href="#contact" class="nav-link d-lg-none">Contact</a>
  <button class="cart-pill btn p-0" data-bs-toggle="offcanvas" data-bs-target="#cartCanvas" aria-label="Open
cart">
    <i class="bi bi-basket3"></i>
    <span class="cart-badge" id="cartCount">0</span>
  </button>
</div>
</div>
</div>
</nav>
```

Category module

```
function productCard(p){
  return `
<div class="col-6 col-md-4 col-lg-3 product-item"
  data-category="${p.category}" data-brand="${p.brand}" data-veg="${p.veg}" data-price="${p.price}">
<div class="product-card">
  <div class="product-thumb">
    <span class="veg-dot ${p.veg ? '' : 'nonveg'}" title="${p.veg ? 'Vegetarian' : 'Non-vegetarian'}"></span>
    ${p.emoji}
  </div>
  <div class="product-body">
    <div class="product-brand">${p.brand}</div>
    <div class="product-name">${p.name}</div>
    <div class="d-flex justify-content-between align-items-center">
      <div class="product-price">₹${p.price} <small>/unit</small></div>
    </div>
    <div class="d-flex gap-2 mt-3">
      <button class="btn btn-sm btn-outline-forest flex-grow-1 rounded-pill"
onclick="openProductModal(${p.id})">View</button>
      <button class="btn btn-sm btn-tomato rounded-pill" onclick="addToCart(${p.id},1)" aria-label="Add
${p.name} to cart"><i class="bi bi-plus-lg"></i></button>
    </div>
  </div>
</div>
```

Screenshots

Homepage



harvestly. Home Categories Offers Shop Contact

DELIVERED IN 19 MINUTES, ON AVERAGE

Farm to door, *before your coffee's cold.*

Harvestly picks, packs and delivers groceries from local farms and trusted brands — fruits still cool from the crate, milk that hasn't left the chain.

[Start shopping](#) [Browse categories](#)

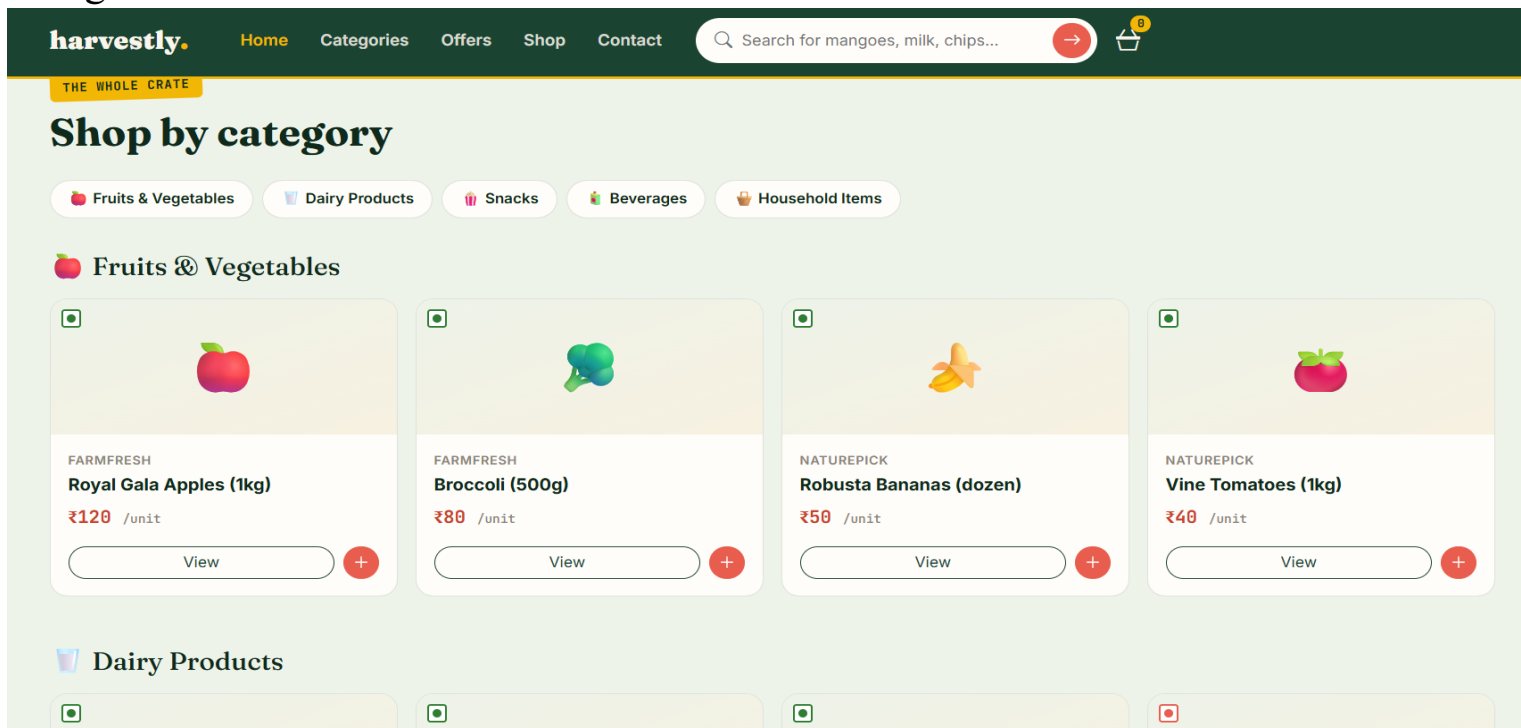
4.8★ 120+ 19 min
12k+ ratings local farms avg. delivery

NEXT SLOT ARRIVES IN
11:43
Order now for the next dispatch window

HANDPICKED TODAY

Featured grocery items

Categories



harvestly. Home Categories Offers Shop Contact

THE WHOLE CRATE

Shop by category

Fruits & Vegetables Dairy Products Snacks Beverages Household Items

Fruits & Vegetables

 FARMFRESH Royal Gala Apples (1kg) ₹120 /unit View	 FARMFRESH Broccoli (500g) ₹80 /unit View	 NATUREPICK Robusta Bananas (dozen) ₹50 /unit View	 NATUREPICK Vine Tomatoes (1kg) ₹40 /unit View
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Dairy Products

Offers

DON'T MISS OUT

This week's offers

COMBO DEAL

Breakfast Combo — milk + bread + eggs

Bundle your morning essentials and save ₹60 compared to buying separately.

Shop By Category

FULL CATALOG

Shop & filter everything

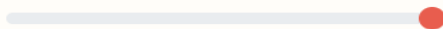
FILTERS

Clear all

DIET

- ☐ All
- ☐ ☒ Vegetarian
- ☒ ☐ Non-Vegetarian

PRICE RANGE — UP TO ₹350



CATEGORY

- ☒ Fruits & Vegetables
- ☒ Dairy Products
- ☒ Snacks
- ☒ Beverages
- ☒ Household Items

BRAND

- ☒ Amul
- ☒ Britannia

2 products found



COUNTRYNEST

Farm Eggs (12 pcs)

₹80 /unit

View



MCCAIN

Chicken Nuggets (400g, frozen)

₹250 /unit

View





**CHANDIGARH
UNIVERSITY**
Discover. Learn. Empower.

**NAAC
GRADE A+**
Accredited University

Contact

harvestly.
Grocery delivery built around freshness and speed — sourced from local farms, delivered to your door.
[f](#) [@](#) [X](#)

SHOP
[Categories](#)
[Offers](#)
[Full catalog](#)

COMPANY
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STAY FRESH — GET OUR NEWSLETTER

[Join](#)
📍 Patiala, Punjab, India · 📞 1800-123-4567

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Learning Outcomes

1. Learned to develop a responsive website using HTML5 and CSS3.
2. Understood the use of the Bootstrap Grid System for responsive layouts.
3. Gained experience in creating navigation bars, cards, carousels, and modals using Bootstrap 5.
4. Implemented dynamic features such as product filtering and shopping cart functionality using JavaScript.
5. Learned DOM manipulation and event handling in JavaScript.
6. Designed ER and DFD diagrams for system analysis and database planning.